RDBMS - SQL
Oracle
SQL Server

1971
Dr. E.F Codd

ACID properties
SQL Server
Oracle
PostgreSQL

Distrubuted-SQL

Distributed
But Structured & ACID
complaint but not full

Yugabyte, CockroachDB

NoSQL
Distributed
Not Structured
No ACID compliance
No Relational Model

MongoDB
CouchBase

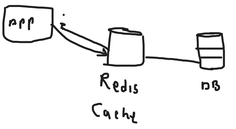
NoSQL Classification

1) Key-Value Pair
2) Document Database
3) Wide Column Stores
4) Graph

Key-Value

Key(001) --> Data(sami, trainer,hyderabad)

BerkleyDB, Redis DB



Document Stores

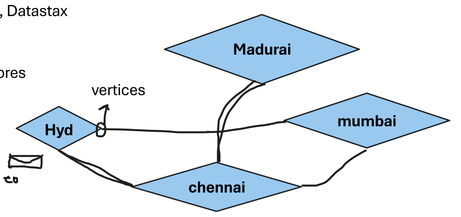
{ Name: sami, designation: Trainer, location: Hyderabad }
{ Name: Raj, designation: Salesman }

MongoDB, Couchbase

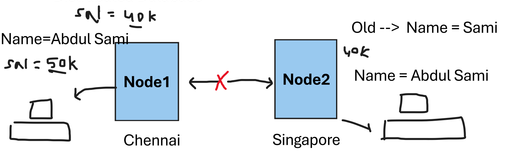
Wide Column Stores

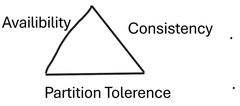
Cassandra, Datastax

Graph Stores



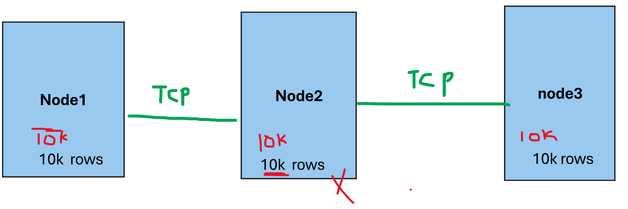
Distrubuted Database





Availability & Partition Tolerence (AP)
Consistency & Partition Toelrence (CP)

Table: Sales with 30k rows



Client Programs

1) CLI Mongo Shell (mongosh) (mongo)
2) Compass (GUI Client)

Third Party GUI Clients:

DBeaver (professional Version)
NoSQL Manager
SQLYog

16GB RAM

50% of (16GB - 1GB)

(16-1)/2
15/2 = 7.5 GB

2GB RAM

2-1 / 2 = 1/2 = 512MB